The effectiveness of a custom AI chatbot for T2DM health literacy: An evaluation study

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The effectiveness of a custom AI chatbot for T2DM health literacy: An evaluation study

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Abstract

Background: People living with chronic diseases are increasingly seeking health information online. For individuals with diabetes, traditional educational materials often lack reliability and fail to engage or empower them effectively. Innovative approaches, such as Retrieval-Augmented Generation (RAG) powered by large language models (LLMs), have the potential to enhance health literacy by delivering interactive, medically accurate, and user-focused resources based on trusted sources.

Objective: To evaluate the effectiveness of a custom RAG-based AI chatbot designed to improve health literacy in type 2 diabetes mellitus (T2DM) by sourcing information from validated reference documents and attributing sources.

Methods: A T2DM chatbot was developed using a fixed prompt and reference documents. Two evaluations were performed: 1) a curated set of 44 questions assessed by a specialist for appropriateness (appropriate, partly appropriate, or inappropriate) and source attribution (matched, partly matched, unmatched or general knowledge); 2) a simulated consultation of 16 queries reflecting a typical patient's concerns.

Results: Of the 44 evaluated questions, 32 responses cited reference documents and 12 were attributed to general knowledge. Among the sourced responses, 30 (94%) were deemed fully appropriate, with the remaining 2 partly appropriate. Of the 12 general knowledge responses, 1 was inappropriate. In the 16-question simulated consultation, all responses were fully appropriate and sourced from the reference documents.

Conclusions: A RAG-based LLM chatbot can deliver contextually appropriate, empathetic, and clinically credible responses for T2DM queries. By consistently citing trusted sources and notifying users when relying on general knowledge, this approach enhances transparency and trust. The findings have relevance for health educators, highlighting that patient centric reference documents—structured to address frequent patient questions—are particularly effective. Moreover, instances where the chatbot signals that it has drawn on general knowledge, can provide opportunities for health educators to refine and expand their materials, ensuring that more future queries are answered from trusted sources. The findings suggest that such chatbots may support patient education, promote self-management, and be readily adapted to other health contexts.

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Original Manuscript

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Keywords: conversational agent; chatbot; ChatGPT; LLM; RAG; T2DM; diabetes; health literacy; clinical credibility; trust

Introduction

Overview

Diabetes is a highly prevalent chronic disease, with an estimated 537 million people worldwide living with diabetes (1). Type 2 diabetes mellitus (T2DM) accounts for approximately 90% of this number. Diabetes is associated with a number of serious complications, and is the most common cause of: kidney failure requiring dialysis, lower limb amputation, and blindness in working-age adults. It is also associated with an increased risk of heart disease (twice the risk of ischaemic heart disease), stroke (twice as common in people with diabetes), and all-cause mortality (2).

Diabetes is characterized by an increase in glucose levels in the blood, which occurs due to inability of the pancreas, through a combination of genetic and environmental factors, to produce enough insulin to meet the body's demands (3). It is well-established that treatment of diabetes, through optimal control of blood glucose, blood pressure and serum cholesterol, is associated with a significant reduction in the development of complications, disease burden, and mortality (4). This is hugely beneficial both from an individual perspective, with reduced morbidity, mortality and increased quality of life, but also from an economic perspective, as treatment of complications, rather than preventive care, accounts for the vast majority of cost associated with diabetes care.

Diabetes education and self-management is acknowledged as the cornerstone of effective diabetes management in the major international consensus guidelines from the American Diabetes Association, the European Association for the Study of Diabetes, and the World Health Organization (5). By equipping individuals with the knowledge and tools necessary to understand their condition, education empowers them to engage in effective self-care, reducing the risk of complications (5). Health literacy plays a key role in this (6–8). Health literacy is defined as the ability to access, interpret, and evaluate health information to make informed decisions about care (9). However, low levels of health literacy remain a critical challenge (10). Traditional approaches to health literacy, such as informational booklets provided by healthcare professionals, often fall short. These materials typically lack interactivity, fail to offer opportunities for personalized inquiry, and do little to cultivate the critical thinking and autonomy essential for effective chronic disease management (11).

It has been proposed that improving health literacy in T2DM management requires innovative solutions, including the use of simplified and interactive educational tools (12,13). This paper evaluates an AI-driven conversational agent chatbot, to enhance health literacy in T2DM by providing interactive, accessible, and personalized guidance grounded in evidence-based medical documents.

Background

The use of chatbot technology, leveraging artificial intelligence (AI) and natural language processing (NLP), is gaining significant attention as a means to enhance interactive patient engagement (14,15). These chatbots offer several advantages, including human-like interactivity with a knowledge base and autonomous access, addressing the need for dynamic, question-driven support to individuals managing their own health whilst living with a chronic disease.

Conversational agents have been successfully implemented in a variety of healthcare settings, providing both educational and supportive roles in chronic disease management (16,17). Public acceptance of these tools continues to grow, particularly with the advent of powerful generative AI models like ChatGPT, which play an acknowledged role in public health education by offering information and answering queries related to health promotion and disease prevention (12).

Generative Pretrained Transformers (GPTs) in general, excel in providing conversational flexibility and autonomy in querying, making them promising tools for improving health literacy (18). ChatGPT specifically, has been applied to patient education in T2DM (14,19). For example, in a study evaluating responses to T2DM self-management questions, clinicians deemed ChatGPT's answers appropriate in 98.5% of cases, rating its reliability as superior to standard search engines (14). However, concerns remain about the potential for erroneous or harmful responses (16,20) and the general lack of transparency regarding the source of information provided by ChatGPT. This limitation perpetuates the reliance on unverified medical information that is commonly encountered during web-based information-seeking (13), with online resources failing to meet the needs of patients (21), including those with T2DM (22).

ChatGPT and similar models are fine-tuned large language models (LLMs) designed to respond to prompts and user conversations in a question-and-answer format. These models operate across diverse tasks without requiring additional fine-tuning, emphasizing the importance of effective prompt engineering—structuring prompts to guide chatbot behavior effectively. This approach aligns with the "pre-train, prompt, and predict" paradigm (23), in which models are pre-trained on large text corpora to produce textual responses, whilst being guided by a prompt to make the text output relevant for the task at hand.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) offers a solution to the issue of medical credibility by anchoring LLM responses to specific, reference documents (24,25). The overall architecture of the RAG LLM is shown in Figure 1. The user query is combined with the prompt to provide an input to the system. The prompt that guides the LLM should be pre-designed for the T2DM health literacy chatbot task. Medical reference documents are indexed to extract the lexical meaning (embeddings) of the text. The Retrieval stage matches the meaning of the query and prompt with the documents in order to retrieve relevant chunks of text. The LLM itself is pre-trained on billions of text samples over diverse topics. The pre-trained LLM processes the relevant document text chunks, together with the prompt and query to formulate the response. All queries and responses from the session are provided to the LLM so the historical context of the conversation is taken into account by the LLM when responding to new queries.

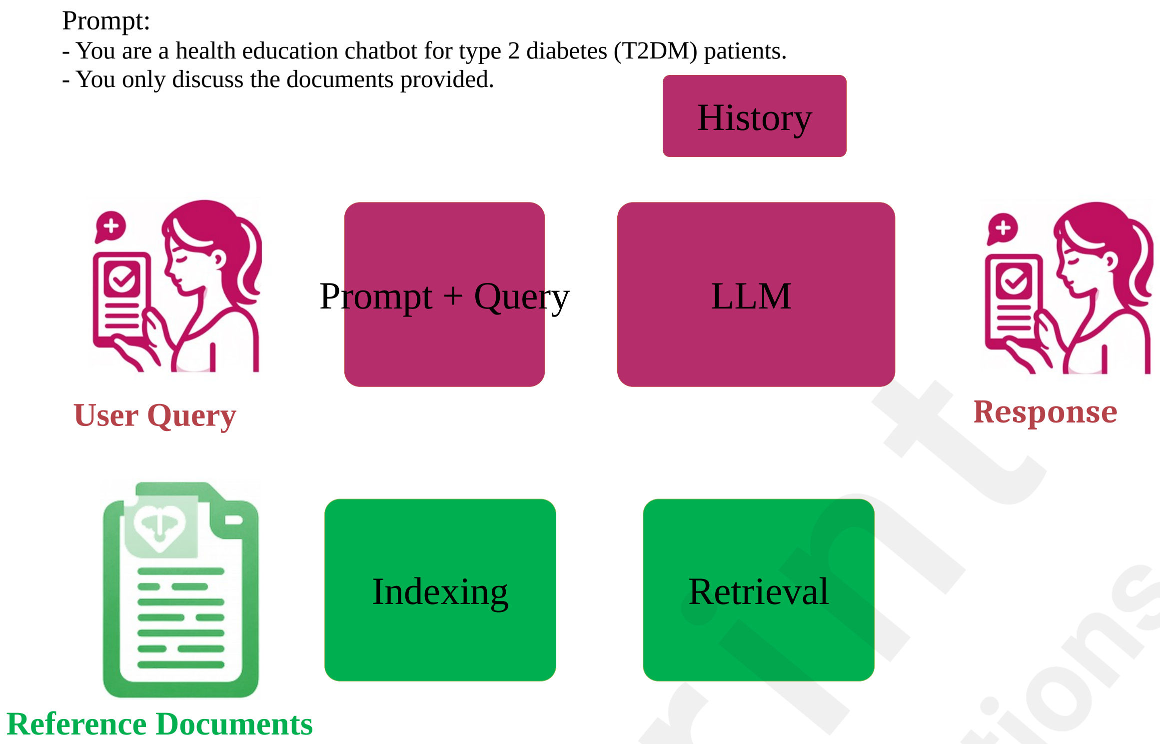

Figure 1 Retrieval-Augmented Generation (RAG) LLM Architecture. *The LLM responds to a combined prompt and query. The prompt is designed to control the LLM response, whilst the query is entered by the user. The LLM draws upon relevant information retrieved from the reference documents to respond to the query. The History of queries and responses is provided to the LLM to retain the historical context of the conversation in the session.*

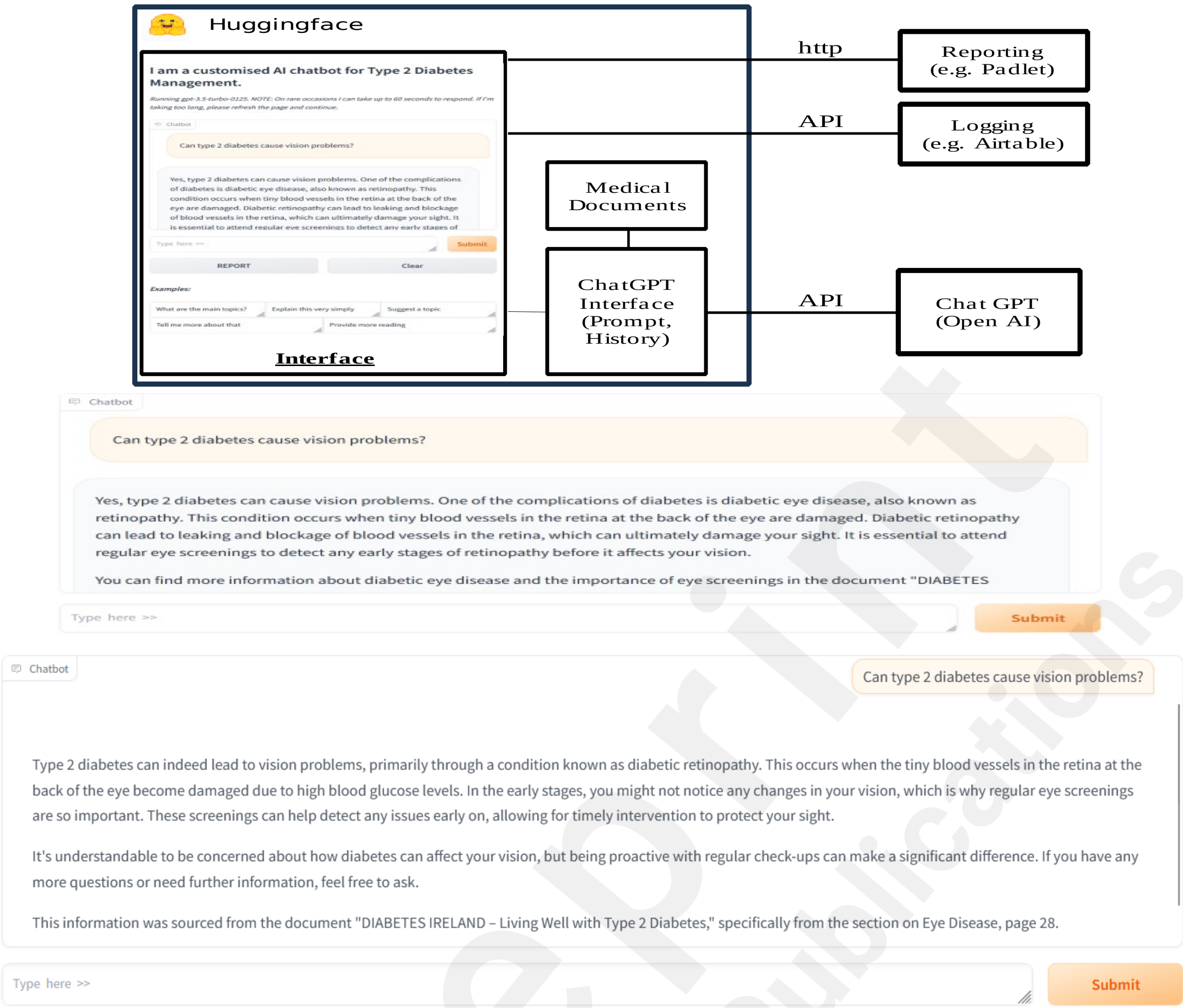

Figure 2 Chatbot system architecture. *Shows the chatbot, implemented in the Python programming language hosted on the Huggingface Hub platform. The cloud based system, hosts the user interface and implement the RAG LLM functionality, communicating over Application Programming Interfaces (APIs), with other cloud based services, Open AI's ChatGPT, Airtable for date/time logging, and Padlet for reporting.*

A more detailed description of the operation of the RAG aspect is shown in Figure 3, illustrating the three main steps involved in RAG: i) Indexing, ii) Retrieval, iii) Generation (24,25).

Before retrieval can occur, documents must first be indexed to enable efficient searching and matching based on semantic similarity. This step further involves 'Document Preprocessing', where the reference documents are segmented into smaller chunks, and formatted appropriately; 'Embedding Generation', where each document chunk is transformed into a numerical vector representation using OpenAI's embedding model or equivalent; 'Vector Storage', where the generated embeddings are stored in a vector database, allowing for efficient similarity-based searches.

Retrieval involves searching and selecting relevant sections from the reference documents in the vector store by comparison with the vector embedding of the search query. The retrieved chunks are ranked according to relevance scores, where the highest-ranked document chunk (Chunk 1) has the highest similarity score to the query. The second-ranked document chunk (Chunk 2) follows as the next most relevant, and so forth up to k documents, where k is typically four by default. The ranking aims to ensure that the top results provide the best contextual grounding for the generation phase.

Once the relevant chunks are retrieved, the highest-ranked chunks are appended to the input prompt and given to the LLM along with the user query. In this Generation phase, the LLM generates a response based on the query, prompt, retrieved information and its internal knowledge. This also helps overcome hallucination issues in large language models by grounding the response in factual, reference documents.

Interested readers are referred to (24,25) for more technical and expansive details of RAG.

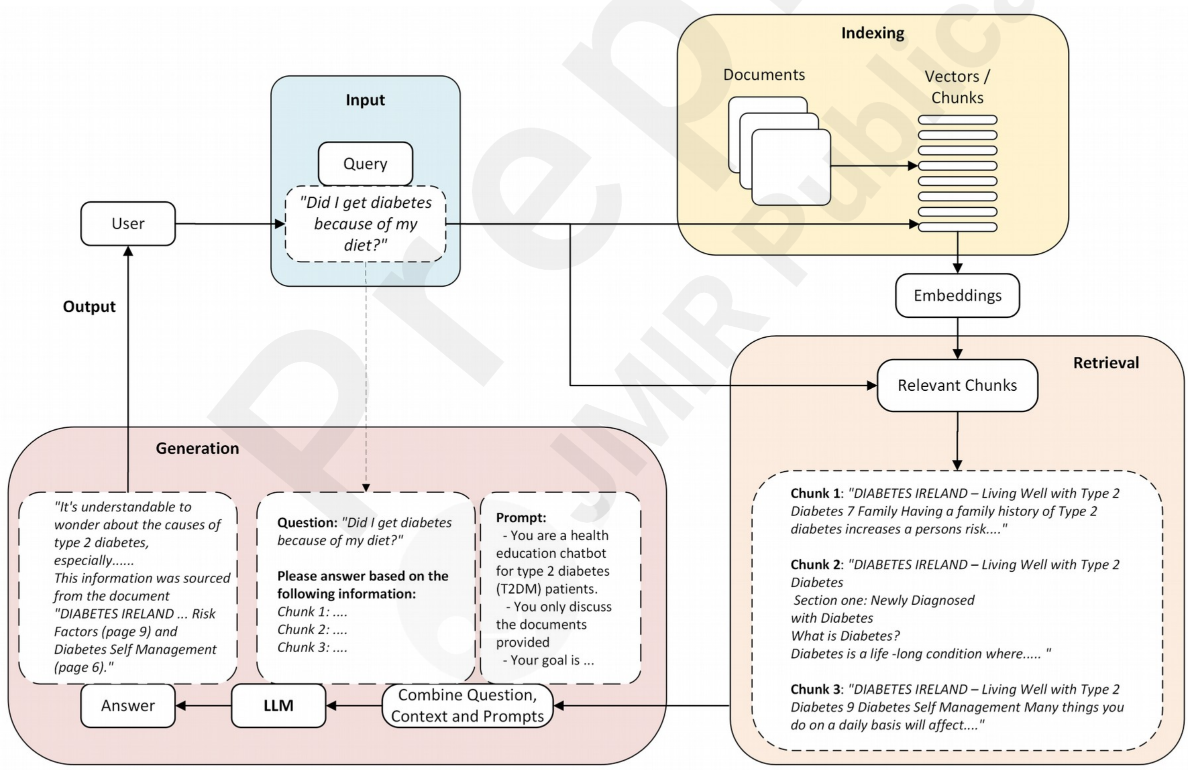

Figure 3 A detailed illustration of the RAG operation. *Illustrates the three steps of i) Indexing; in which reference documents are split into chunks and stored in a vectorstore for later retrieval; ii) Retrieval: in which the top-k relevant chunks are retrieved based on semantic similarity to the query; iii) Generation: in which the Prompt and Query are combined with the Retrieved chunks and sent to the LLM to output a response. (Based on a diagram in [25])*

Drawing on chunks of relevant data retrieved from the reference documents, as discussed, RAG provides LLMs with the ability to attribute the source of the information in the response. It is apparent from Figure 3, that the retrieved chunks are large enough, so they typically contain the page and document header information that is included on the document pages. The feasibility of attributing the source in this way, with suitable prompting, has been demonstrated by the authors (26).

Although the feasibility of RAG-based question-and-answer chatbots to answer queries appropriately and to attribute the response to the medical source has been demonstrated (26), several issues remain to be addressed. One of these is to appropriately answer queries that are not easily found in the reference documents, whilst also preserving medical credibility. The feasibility study (26) showed responses that were judged to be partly appropriate or inappropriate were associated with the chatbot not easily finding directly relevant information in the documents provided to the LLM. This resulted in inappropriate or partly appropriate synthesis of various information in the document and the LLM resorting to its general knowledge in order to respond to the query. Another issue related to the manner in which the chatbot notified the user that it is resorting to its general knowledge for the response, and not the provided reference documents. Responding clearly in this situation is important for allowing flexibility in responses, whilst also warning users about medical validity of the responses.

The LLM outputting source citation information and guidance on clinical validity, addresses the issue of medical credibility that other work fails to address (27,28). In this present evaluation study, the drawbacks of the feasibility study (26) have been addressed through revised prompting and updating the RAG documents to be more easily queried by the LLM and are therefore more suited to RAG approaches. In addition, recent advancements in LLM models have been incorporated (OpenAI's gpt-4o-mini). Additionally, a more comprehensive evaluation methodology has been adopted, involving the evaluation of 44 questions curated by a specialist, and a simulated patient conversation.

Methods

Design and architecture

The system (Figure 2), was designed around a custom conversational agent chatbot (26), in order to address several limitations associated with the public OpenAI chatbot interface, particularly with regard to prompt control, user privacy, and the relevance of source material. To ensure controlled interactions, the chatbot was programmed with a fixed, carefully constructed prompt that guided all responses. This approach prevented user modifications and minimized the risk of inappropriate outputs that may result from unsuitable user prompts or malicious prompt attacks. The chatbot's knowledge base was constrained to validated documents, ensuring that the chatbot delivered medically accurate and contextually relevant information while reducing reliance on the language model's general training data. Additionally, the prompt was engineered to request that the LLM attribute the source of the response information to the specific section of the medical documents it was sourced from. It was asked to highlight if information was not sourced from the documents, to ensure transparency.

User privacy was ensured by employing a custom interface that dissociated queries from individual accounts. The interface used shared access credentials rather than personalized logins, thus ensuring anonymity. The system's backend was developed in Python, utilizing Gradio libraries for the user interface. The secure API ensured private key management. The chatbot was made accessible via a secure URL, allowing participants to interact through web browsers on any device. Interaction data was anonymized and logged in an external database for the purpose of evaluating usage, with no sensitive user data or chat information being stored.

Large language models and prompts

In the standard paradigm of supervised machine learning, a prediction model is typically trained on training examples and optimized according to a cost function so that predictions on new data closely match the characteristics of the training data. Therefore, the tasks carried out by these models are generally task and domain specific. In deep learning the training process can involve hundreds of hours of compute intensive resources and millions of training examples. LLMs are deep learning models pre-trained on enormous natural language datasets to learn the characteristics of language through the task of predicting the probability of the following textual data. Typically, for a chat model, pre-training is followed by fine-tuning towards suitability for a chat function (23). In contrast to task specific AI models, chat models are fine-tuned LLMs that are designed to respond to prompts and conversation history, incorporating a question-and-answer format. By responding to prompts, the chat LLM can operate on a variety of tasks without additional fine-tuning. This makes prompt engineering (designing the most appropriate prompt for the task), an important aspect of chatbot design (23).

Table 1 The LLM prompt. *Shows the content and structure of the prompt, explaining the effect the prompt is designed to have on the chatbot behavior.*

Prompt	Explanation
- You are a health education chatbot for type 2 diabetes (T2DM) patients. - You only discuss the documents provided.	Constrain responses
- Your goal is to improve health management of chronic diseases. - Your answers should explain things clearly, sensitively and avoid jargon. - Always show empathy in your responses. - Do not appear to blame users for their condition. - Do not start an answer with "Yes" or "Certainly" which conveys certainty. Instead, explain the diverse view points of the answer.	Set the goal and style
- You are allowed to chat with the user in general conversation to support your goal. - If the user goes off topic, gently and politely let them know and go back on topic.	Constrain responses
- You must be safe to use. If you don't know the answer then say that. Do not make anything up.	Reduce hallucinations
- Always finish by establishing where you found the information in the documents provided, including the document, the section and page number. - If you did not find the information in the documents provided, then say "this information was sourced from my general knowledge, so I can't be certain that it's clinically valid."	Provide a source (trust) State source unknown (trust)

In this system the LLM prompt (as shown in Table 1) was designed to ensure the chatbot interacts with the user in the desired way and to prevent the issues encountered in the previous version (26). The changes required to reduce the undesirable responses encountered in the previous version are shown in Table 2, illustrating how the prompt influenced the chatbot responses as the chatbot was developed from its feasibility (26) to the present study. The prompt starts by setting the persona of the LLM and constraining the chatbot to only answer questions relating to the documents provided to the LLM. The wording of this part of the prompt allows the chatbot to discuss limited information related to the documents. Previously (Table 2), the LLM was allowed to answer related questions. Although this allowed flexibility and autonomous user enquiry, it resulted in the LLM straying away from the documents on several occasions during testing. With the present prompt, the LLM tends to avoid this but can still allow some range of conversation due to the goal constraint in the next part of the prompt.

The next section of the prompt sets the goal and style. The LLM's goal is set here, and it is directed to output responses that show empathy and avoid language that appears to blame the user for their condition. If this is not included the responses can come across as purely factual. For example, the question *"Could I have prevented diabetes if I ate better and exercised"* was answered as: *"Yes, making healthy lifestyle choices…"*, in the previous prompt. The present prompt causes the response to be more empathetic in style: *"It's understandable to wonder about the role of diet and exercise in the development of type 2 diabetes. While a healthier diet and regular physical activity…"*.

The next section the prompt directs the chatbot to re-direct the conversation if it strays off topic, outputting text such as: *"That question is a bit off-topic. Let's focus on discussing Type 2 diabetes and how it can affect your daily life. If you have any questions related to diabetes or the information provided in the documents, feel free to ask!"*. This is followed by a section that directs the LLM to avoid hallucinating a response, and to answer instead, that it does not know the answer to the query.

The last two sections in the prompt direct the LLM to output the location of the source of the answer. In cases where the answer does not exist in the source the chatbot may use its general knowledge to answer the query. When this happens the LLM is directed to say: "*this information was sourced from my general knowledge, so I can't be certain that it's clinically valid* ". The statement about certainty and clinical validity was added to make it clear to users that this information is not as clinically credible as information sourced from the reference documents. This allows the flexibility for users to ask a variety of questions where the answer may not easily be determined from the documents, whilst allowing them to assess the credibility of the responses.

Table 2 Prompt development. *Illustrates how the prompt developed to constrain the chatbot to answer queries primarily from the reference documents and to clearly alert users if general knowledge was utilized to answer queries that were not readily sourced from the reference documents.*

Present Prompt	Previous Prompt	Explanation
- You are a health education	- You are a health education	Additional

chatbot for type 2 diabetes (T2DM) patients. - You only discuss the documents provided.	chatbot for type 2 diabetes (T2DM) patients. - You only discuss the documents provided and information related to them.	constraints to source within the reference documents.
- Your goal is to improve health management of chronic diseases. - Your answers should explain things clearly, sensitively and avoid jargon. - Always show empathy in your responses. - Do not appear to blame users for their condition. - Do not start an answer with "Yes" or "Certainly" which conveys certainty. Instead, explain the diverse view points of the answer.	- Your goal is to improve health management of chronic diseases. - You also need to help users explore their attitudes to their health in relation to T2DM. - Your answers should explain things clearly and avoid jargon. - You are allowed to chat with the user in general conversation to support your goal.	Setting the style to display empathy.
- You are allowed to chat with the user in general conversation to support your goal.	- You are allowed to chat with the user in general conversation to support your goal.	Limited scope for general conversation. No change.
- If the user goes off topic, gently and politely let them know and go back on topic.	- If the user goes off topic, gently and politely let them know and go back on topic.	Staying on topic. No change
- You must be safe to use. If you don't know the answer then say that. Do not make anything up.	- You must be safe to use. If you don't know the answer then say that. Do not make anything up.	Reduce hallucinations. No change
- Always finish by establishing where you found the information in the documents provided, including the document, the section and page number. - If you did not find the information in the documents provided, then say "this information was sourced from my general knowledge, so I can't be certain that it's clinically valid."	- Always finish by establishing where you found the information in the documents provided, including the document, the section and page number. - If you did not find the information in the documents provided, then say "this information was sourced from my general knowledge."	Establish the source for medical credibility. A small change to the used. to clarify the response is not from the reference documents.

Experimental Setup

The RAG LLM is designed to answer patient queries in an interactive way based on medically credible sources. Motivated by a desire to replicate the type and quantity of documents a newly diagnosed patient might be advised to consult for health literacy, the chatbot's knowledge base consisted of two health literacy documents, A third, practitioner oriented document was included for testing to understand what effect its inclusion would have on responses. To facilitate document source information retrieval, each document contained a document name and page number on each page. The first document was a patient information booklet on T2DM, designed to provide structured and patient-friendly answers, reflecting the types of queries typically encountered by individuals managing T2DM (29). The second document was a guide for healthcare professionals, which contained detailed but less structured content aimed at medical practitioners (30). The third was a primary care clinical practice document for T2DM management, which provided clinical management guidance in a structured way (31). The inclusion of these varied documents enabled the evaluation of the chatbot's ability to source and output information appropriate for patient-centric interactions while addressing the challenges posed by more complex knowledge sources.

The document indexing and retrieval chain (Figure 3), was configured with a chunk size of 5000 so that it was likely that the retrieved data would include page and document information that appears on each page. The retrieval was configured to retrieve the top 4 documents (the default setting).

Two evaluations were conducted; one involved a series of curated questions related to diabetes self-management, employing a similar methodology to Hernandez et al. (14). The second involved a simulated clinical consultation conversation employing a similar methodology to Sng et al. (19).

The chatbot's responses were assessed based on two primary dimensions: i) appropriateness and ii) source attribution. Appropriateness was classified into three categories: a) appropriate (the answer is fully appropriate); b) inappropriate (the answer is not suitable) or; c) partly appropriate (the answer has aspects that may be inappropriate). Responses may be deemed to be Partly Appropriate due to facts or style. Source attribution was evaluated by determining whether the response was correctly linked to one of the three documents in the knowledge base, according to four categories: a) correctly matched to the source (matched); b) incorrectly matched (unmatched); c) partly matched (meaning the information appears in the source but in a different location); d) attributed to the LLM's general knowledge (general). The source attribution was matched according to whether the reviewer found the associated information on the page cited. In situations where this was not readily apparent, the RAG Chunks (as illustrated in Figure 3), were examined to verify the source.

Question Evaluation:

For question evaluation, the chatbot was asked questions related to diabetes self-management, employing a similar methodology and questions to Hernandez et al. (14). Having selected appropriate questions for the context and country (Ireland), the responses were recorded and reviewed by a clinical review team of two specialist endocrinologists on the appropriateness dimension. The clinical review team consisted of a highly qualified and experienced consultant endocrinologist and an experienced specialist endocrinology registrar. A consensus opinion was used. The Content column shows the appropriateness categorized into the three defined categories (Table 3). The source attribution of the questions was evaluated by a researcher and categorized into the four defined categories in the Source column (Table 3). The source column lists the status of the evaluation and also which document it was sourced from. For example: Matched (Patient) means the source was correctly matched to the patient booklet document. The word "General" in this column means the chatbot stated "*this information was sourced from my general knowledge*.". The 70 questions arrived at by Hernandez et al. (14) were reviewed by the specialist to remove questions that were not deemed relevant to the local population, resulting in 44 most relevant questions. Some examples of removed questions were "Does diabetes cause hair loss?" and "Should I avoid caffeine if I have type 2 diabetes?". These are not questions that are typically encountered by the specialist in Ireland. Due to the observed consistency of the answers to repeated questions, each question in the list was asked once.

Consultation Evaluation:

To simulate a clinical consultation, a clinical review team of two specialist endocrinologists posed a free-flowing series of questions to the chatbot, reflecting a typical interaction with a newly diagnosed T2DM patient, employing a similar methodology to Sng et al. (19). The clinical review team consisted of a highly qualified and experienced consultant endocrinologist and an experienced specialist endocrinology registrar. A consensus opinion was used. Questions focused on topics related to diabetes self-management. The chatbot's responses were recorded for subsequent analysis.

The clinical review team posed a series of questions (detailed in Table 8), with the chatbot's responses evaluated for both accuracy and source attribution as described previously. A researcher independently reviewed the source of each response to determine its alignment with the intended document. For example, "Matched (Patient)" indicated a response correctly sourced from the patient information booklet, while "General" signified reliance on the model's general knowledge.

Response Repeatability:

To assess the repeatability of the LLM outputs, the semantic stability across multiple generated responses was evaluated. This ensures meaning-level consistency whilst allowing minor changes in word choice, sentence structure or punctuation. The evaluation involved querying the LLM 20 times for each of the 44 questions. SBERT embedding cosine similarity was used to quantify meaning-level consistency. Cosine similarity score ranges from +1 to -1, where +1 indicates perfect similarity, 0 indicates no similarity and -1 indicates perfect dissimilarity. Typically, SBERT cosine similarity scores are limited in range from 0 to 1. The use of SBERT, which uses an alternative embedding model to Open-AI's embeddings used by the LMM, brings a level of independence to the evaluation.

SBERT embeddings employ a sentence-level representation method optimized for semantic similarity. SBERT transforms responses into dense vector representations, capturing contextual meaning beyond lexical overlap. By computing cosine similarity between these embeddings, the semantically consistency of responses can be ascertained, even if they exhibit lexical variation. SBERT embeddings allow for paraphrase detection and improved robustness to minor rewordings. In Semantic Textual Similarity tasks, human-labelled pairs with near-identical semantics (e.g., paraphrases or reworded sentences) commonly achieve cosine similarity scores of around 0.85 and above when using SBERT embeddings.

Query Difficulty:

It is interesting to evaluate whether the retrieval stage finds more relevant information in the documents, for some questions versus others. To evaluate this the SBERT embedding cosine similarity was measured for the most relevant (top-1) chunk returned by the retriever, compared with the question, for each of the 44 questions. The top-1 was selected because this is the most relevant data retrieved.

Results

Question Evaluation:

The categorized responses to the question evaluation are listed in Table 3 and depicted graphically in Figure 4.

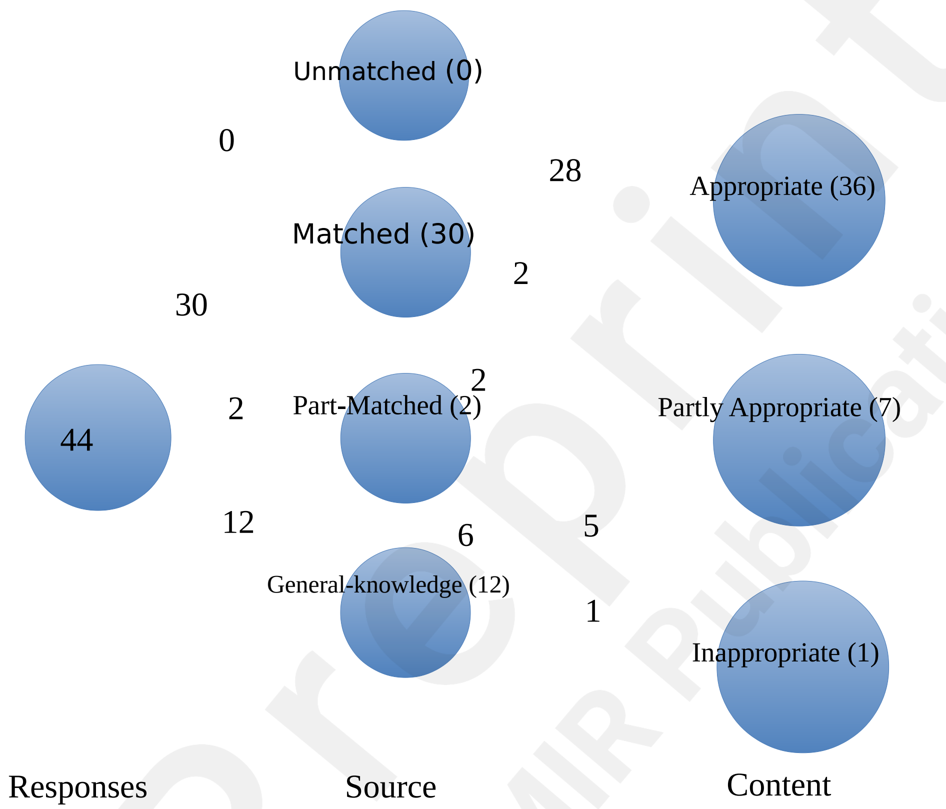

Figure 4 Graphical representation of Question Evaluation. *Shows that all responses were matched to the cited source (either fully or partly), or was cited to have come from the LLMs general knowledge.*

In terms of matching responses to the documents, 32 responses (73%) were sourced from the Reference documents whilst 12 (27%) were not. Of the 32 responses, 30 were matched and 2 were part-matched (meaning the information was in the documents, but the page / section referred to was incorrect). In 12 cases the responses were cited as coming from the LLMs general knowledge (General). In no case, was a cited source found to be unmatched.

Evaluating content, out of 44 responses, 36 (82%) were evaluated as fully appropriate. Seven responses (16%) were evaluated as partly appropriate. One (2%) was evaluated as inappropriate. The fully appropriate responses were judged to be correct in content and the topic was seen to be suitably communicated to the patient. The vast majority of the 36 Appropriate responses were sourced from the Reference documents (30, 83%), whilst 6 Appropriate responses (17%) came from the LLMs General knowledge.

The Partly Appropriate responses (7) were not incorrect, but may have omitted a relevant detail, or mis-stated a fact, or may not have been consistent with how a topic should be communicated. For example (Table 4), the response to a question about memory problems (Q.22), correctly states the details, but fails to mention that this is not very likely, which could lead to unnecessary worry. The response to a question about ketoacidosis (Q.13) appears to match the reference document text, but was found to conflate two different conditions. The vast majority of the 7 Partly Appropriate responses were sourced from the LLMs General knowledge (5, 71%), and therefore finish with the disclaimer around clinical validity. Reference documents were matched in 2 cases: question 13 about ketoacidosis and question 20 related to weight loss. Question 20 (Table 4) referring to weight loss conveys the details from the reference documents accurately, but fails to emphasise that weight loss occurring unintentionally, is unhealthy. Question 13 (Table 4) referring to ketoacidosis conflates two conditions (non-ketotic hyperglycaemia, diabetic ketoacidosis) due to the construction of the sentence in the reference document not being properly interpreted by the LLM “may not always be accompanied by ketones, when onset is gradual (non-ketotic hyperglycaemia)" . The LLM does not notice the meaning of the brackets as identifying non-ketotic hyperglycaemia as the context.

The inappropriate response was judged to have omitted important details related to context. In response to the question “Is an artificial pancreas the same as an insulin pump?”, the response fails to inform the user that this question is not relevant for T2DM, and is therefore judged as inappropriate. Since the response came from the LLM’s general knowledge, it was also somewhat out of date factually.

Table 3 Question Evaluation. *An evaluation of 44 questions posed to the T2DM chatbot. Judged as Appropriate in 80% (35/44) of all responses and 94% of responses that cited a source (30/32).*

Question	Evaluation	
	Content	Source
1. Can type 2 diabetes be prevented?	Appropriate	Matched (IDF)
2. Is type 2 diabetes a lifelong condition?	Appropriate	Matched (Patient)

3.	Can diabetes (DMT2) be inherited?	Appropriate	Matched (Patient)
4.	Can stress affect my blood sugar levels?	Appropriate	Matched (Patient)
5.	Can type 2 diabetes cause frequent urination?	Appropriate	Partly Matched (Patient)
6.	Can type 2 diabetes cause dry mouth?	Partly Appropriate	General
7.	Can type 2 diabetes affect my vision?	Appropriate	Matched (Patient)
8.	Does type 2 diabetes increase the chance of having a heart attack?	Appropriate	Matched (Patient)
9.	Can type 2 diabetes affect my kidney?	Appropriate	Matched (Patient)
10.	Can type 2 diabetes affect my nerves?	Appropriate	Matched (Patient, ICGP)
11.	Can type 2 diabetes cause sexual dysfunction?	Appropriate	Matched (ICGP)
12.	Can type 2 diabetes increase the risk of stroke?	Appropriate	Matched (Patient)
13.	Can type 2 diabetes cause ketoacidosis?	Partly Appropriate	Matched (ICGP)
14.	How is diabetic ketoacidosis diagnosed?	Appropriate	Matched (ICGP)
15.	Can type 2 diabetes affect my fertility or pregnancy?	Appropriate	Matched (Patient, ICGP)
16.	Can type 2 diabetes cause vision problems?	Appropriate	Matched (Patient, ICGP, IDF)
17.	Can type 2 diabetes affect my mental health?	Appropriate	Matched (ICGP)
18.	Can type 2 diabetes cause foot ulcers?	Appropriate	Matched (ICGP)
19.	Can type 2 diabetes cause skin problems?	Appropriate	General
20.	Can type 2 diabetes cause weight loss?	Partly Appropriate	Matched (ICGP)
21.	Can type 2 diabetes cause tingling or numbness in the hands and feet?	Appropriate	Matched (ICGP, IDF)
22.	Can type 2 diabetes cause memory problems?	Partly Appropriate	General
23.	Can type 2 diabetes cause delayed wound healing?	Appropriate	General
24.	Can type 2 diabetes cause dental procedure complications?	Appropriate	General
25.	Should I check my blood pressure regularly if I am diagnosed with type 2 diabetes?	Appropriate	Matched (ICGP, IDF)
26.	Should I have regular eye examinations if I have type 2 diabetes?	Appropriate	Matched (Patient)
27.	Can medications for type 2 diabetes treatment cause hypoglycemia?	Appropriate	Matched (Patient)
28.	Can I get off my diabetic medications if I lose weight?	Appropriate	General
29.	Would having bariatric surgery cure my type 2 diabetes?	Appropriate	Matched (IDF)
30.	How can I lower my chances of a heart attack or stroke if I have type 2 diabetes?	Appropriate	Matched (ICGP)
31.	Is an artificial pancreas the same as an insulin pump?	Inappropriate	General
32.	Can supplements cure my type 2 diabetes?	Partly Appropriate	General
33.	Do I need to follow a special diet for type 2 diabetes?	Appropriate	Matched (Patient)
34.	Can I manage type 2 diabetes through lifestyle changes alone?	Appropriate	Matched (IDF)
35.	Should I avoid food and drinks with sugar?	Appropriate	Matched (ICGP)
36.	Is it necessary to quit smoking if I have type 2 diabetes?	Partly Appropriate	General
37.	Is it necessary to quit alcohol if I have type 2 diabetes?	Appropriate	General
38.	Is it essential to maintain a healthy weight with type 2 diabetes?	Appropriate	Matched (ICGP)
39.	Should I have a yearly flu shot if I have type 2 diabetes?	Appropriate	Partly Matched (Patient)
40.	Should I receive the pneumococcal vaccine if I have type 2	Appropriate	General

diabetes?		
41. How often should I see my primary care provider or endocrinologist if I have type 2 diabetes?	Appropriate*	Matched (ICGP, IDF)
42. What is hemoglobin A1C test?	Appropriate	Matched (ICGP)
43. What level of hemoglobin A1C is considered prediabetic?	Partly Appropriate	General
44. Is it necessary to check my cholesterol levels regularly if I have type 2 diabetes?	Appropriate	Matched (Patient)

NOTE: * : Appropriate for the time the document was written

Table 4 Example Partly Appropriate Responses and evaluation comments. *Demonstrates how the facts and the messaging must be correct in the responses.*

Question	Response (Partly-Appropriate)	Evaluative Comment
13. Can type 2 diabetes cause ketoacidosis?	Type 2 diabetes can indeed lead to a condition called diabetic ketoacidosis (DKA), although it is more commonly associated with type 1 diabetes. DKA occurs when the body starts breaking down fats at an excessive rate, producing ketones, which can lead to a dangerous buildup of acids in the bloodstream. However, it's important to note that DKA in type 2 diabetes is less common and <u>may not always be accompanied by ketones, especially if the onset is gradual. This condition is known as non-ketotic hyperglycaemia.</u> Immediate medical attention is required for DKA, and assessment typically includes a glucometer reading, urinalysis for ketones, and evaluation of hydration status. This information was found in the "IRISH COLLEGE OF GENERAL PRACTITIONERS – A Practical Guide to Integrated Type 2 Diabetes Care," page 30.	"May not always be accompanied by ketones, especially if the onset is gradual. This condition is known as non-ketotic hyperglycaemia", should be removed (there are always ketones in DKA, this seems to be referring to another condition called hyperosmolar hyperglycaemic state.
20. Can type 2 diabetes cause weight loss?	Type 2 diabetes can sometimes lead to weight loss, especially in the early stages of the condition. This can happen for a few reasons. When the body is unable to use insulin effectively, it may start to break down fat and muscle for energy, which can result in weight loss. Additionally, high blood sugar levels can lead to increased urination, causing the body to lose calories and fluids. <u>However, it's important to note that weight loss is not a typical or healthy way to manage type 2 diabetes.</u> Many people with this condition are actually overweight or obese, and managing weight through a healthy diet and regular exercise is often recommended to help control blood sugar levels. If you or someone you know is experiencing unexpected weight loss, it's essential to consult a healthcare professional for proper evaluation and guidance. This information was sourced from the IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care, section 3.3 Diet, and the Irish College of General Practitioners – A Practical Guide to Integrated Type 2 Diabetes Care, Appendix 3.	"However, it's important to note that weight loss is not a typical or healthy way to manage type 2 diabetes. ". This is true for unintentional weight loss, not explained by diet or exercise , and healthy weight loss is discussed accurately in the next sentence. This is difficult to convey at the best of times!
22. Can type 2 diabetes cause memory problems?	Type 2 diabetes can indeed be associated with memory problems. High blood glucose levels over long periods can lead to complications that affect various parts of the body, including the brain. This can potentially impact cognitive functions, such as memory. Additionally, factors like anxiety, depression, and acute illness, which are more	Technically accurate but only over a very long number of years, and only in a very small number of

	prevalent in people with diabetes, can also influence cognitive abilities and memory. It's important to manage blood glucose levels effectively and address any mental health concerns to help mitigate these risks. If you are experiencing memory problems, it would be a good idea to discuss this with your healthcare provider to explore the underlying causes and appropriate interventions. This information was sourced from my general knowledge, so I can't be certain that it's clinically valid.	people. So, accurate, but not a good clinical message as irrelevant (yet very worrying) for most people.

Table 5 Inappropriate Response and evaluation comments. *This response was sourced from LLM general knowledge, showing how LLM knowledge can be become dated. Mainly, it is Inappropriate because it fails to inform the user that the question is not relevant for T2DM.*

Question	Response (Inappropriate)	Evaluative Comment
31. Is an artificial pancreas the same as an insulin pump?	An artificial pancreas and an insulin pump are related but not the same. An insulin pump is a device that delivers insulin continuously throughout the day to help manage blood glucose levels. It requires the user to manually adjust the insulin delivery based on their blood sugar readings, carbohydrate intake, and physical activity. On the other hand, an artificial pancreas is a more advanced system that combines an insulin pump with a continuous glucose monitor (CGM) and a sophisticated algorithm. This system automatically adjusts insulin delivery in real-time based on glucose readings from the CGM, aiming to maintain blood glucose levels within a target range with minimal user intervention. While both devices aim to improve blood glucose management, the artificial pancreas offers a more automated and integrated approach compared to the insulin pump. This information was sourced from my general knowledge, so I can't be certain that it's clinically valid.	Most insulin pumps now act as an 'artificial pancreas'. So, this information has dated. A key point not covered, is that it is not relevant for T2DM.

It is instructive to look at the chunks returned by the Retrieval stage for this inappropriate response that was cited as general knowledge (Table 6) and contrast it to those of an appropriate, cited response (Table 7). It is clear that the retrieved chunks do not have any apparent relevance to the question in the Inappropriate case. Hence the LLM relies on its general knowledge for the response (Table 5), citing that it has done so, as instructed by the prompt. In comparison, the chunks retrieved for an appropriate, cited response (Table 7), are all relevant to the question, and furthermore, page and document information appear in the relevant chunks, enabling the source citation.

Table 6 Retrieved Chunks for the Inappropriate Response. *The retrieved chunks do not have any apparent relevance to the question. Hence the LLM relies on its general knowledge for this response, citing that it has done so, as instructed by the Prompt.*

Question	Retrieved Chunks (shortened for brevity)	Explanation
31. Is an artificial pancreas the same as an insulin pump?	**Chunk 1:** DIABETES IRELAND – Living Well with Type 2 Diabetes 26 Insulin injections Insulin may be prescribed for you if other treatment options are unsuitable or if they are no longer able to keep your diabetes under control. Insulin may be prescribed in addition to other treatments for Type 2 diabetes. …. You should know and be able to specify the insulin product(s) you are on by brand name and strength as insulin products are not inter -changeable.	Relates to insulin injections, not an insulin pump or artificial pancreas.
	Chunk 2: IRISH COLLEGE OF GENERAL PRACTITIONERS – A Practical Guide to Integrated Type 2 Diabetes Care | 68 © ICGP February 2016 Appendix 8 If insulin to continue: • Prescribe once daily morning dose of isophane insulin or long-acting insulin Glargine based on 25% less than total previous daily insulin dose …. Contact the Diabetes Nurse Specialists or Palliative Care Team if advice required.	Relates an end of life diabetes management care pathway diagram, not an insulin pump or artificial pancreas.
	Chunk 3: and empowered to treat hypoglycaemia and all patients on insulin should be supplied with a Glucagon Hypokit and be familiar with its use. Insulin initiation is usually carried out in a diabetes day care centre; however Insulin Initiation in General Practice is considered feasible once the practice has established its integrated diabetes service, including education of the Practice Nurses and GPs and the availability of the Community Diabetes Nurse Specialists.	Does not refer to an insulin pump or artificial pancreas.
	Chunk 4: IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 201728 Many guidelines consider basal insulin as the best option when the HbA1c target has not been achieved… Since the completion of the review of guidelines for this document, new cardiovascular safety trials have been reported with two SGLT2 inhibitors (empagliflozin and canagliflozin) and with two long	Does not refer to an insulin pump or artificial pancreas.

Table 7 Retrieved Chunks for an Appropriate Response. *The retrieved chunks are relevant and include source document page and section information gleaned from the relevant pages of the documents. The full LLM Response is also shown.*

Question	Retrieved Chunks (shortened for brevity)	Explanation
Can I manage type 2 diabetes through lifestyle changes alone?	**Chunk 1:** IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 201717 3.2 Lifestyle changes The variable recommendations between guidelines reflect	Relevant and balanced

	the controversy around starting management of the patient with T2D with lifestyle changes alone and postponing medication, which would be started only in those cases in whom lifestyle changes were not sufficient to achieve the metabolic targets. …	
	Chunk 2: DIABETES IRELAND – Living Well with Type 2 Diabetes 9 Diabetes Self Management Many things you do on a daily basis will affect your blood glucose levels, so it is very important that you take an active part in the management of your diabetes… This includes: • Eating a healthy diet • Doing regular physical activity	Relevant. Includes lifestyle advice.
	Chunk 3: IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 201715 Lifestyle changes	Refers to a heading on an illustrative page. No relevant content.
	Chunk 4: IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 201719 3.4 Physical activity All guidelines consider physical activity or exercise as part of lifestyle modifications. Some specify moderate aerobic physical activity such as walking for at least 150 minutes per week at intervals of no longer than 48 hours….	Relevant. Includes lifestyle advice.
	Response	
Can I manage type 2 diabetes through lifestyle changes alone?	Managing type 2 diabetes (T2DM) through lifestyle changes alone is a topic with varying viewpoints. According to the IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care, there is some controversy around this approach. Some guidelines suggest starting with lifestyle changes alone and postponing medication, only introducing it if lifestyle changes are not sufficient to achieve metabolic targets (IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 2017, Section 3.2, Page 17). Lifestyle changes can include adopting a healthy diet, increasing physical activity, and losing weight. These changes can significantly improve blood sugar control and overall health. However, the effectiveness of lifestyle changes alone can vary from person to person. Some individuals may achieve good control of their blood sugar levels with lifestyle changes alone, while others may need medication to help manage their condition. It's important to work closely with your healthcare team to determine the best approach for you. They can provide guidance on the specific lifestyle changes that may be most beneficial and help monitor your progress. This information was sourced from the IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 2017, Section 3.2, Page 17.	

Consultation Evaluation:

The question evaluation was designed to evaluate the chatbot with a wide range of queries that have been clinically judged as relevant. However, a conversation between the user and the chatbot allows the user to follow a line of inquiry, that allows an autonomously driven investigation of the reference documents. A simulated conversation is shown in Table 8, in which an expert played the role of the user, in a typical conversation after diagnosis with T2DM.

Table 8 Consultation Evaluation. *An evaluation of 16 free-flow question in a simulated patient conversation. All responses were evaluated as Appropriate and all Matched the reference documents.*

Question	Evaluation	
	Content	Source
I have just been diagnosed with type 2 diabetes and I am very worried	Appropriate	Matched (Patient)
I am worried that I could lose my sight, or end up on dialysis	Appropriate	Matched (Patient)
Can I manage diabetes by just changing my diet	Appropriate	Matched (IDF)
What are the chances that I could lose my sight with diabetes	Appropriate	Matched (Patient)
What are the chances of needing dialysis with diabetes	Appropriate	Matched (ICGP)
What kind of cardiovascular issues are common?	Appropriate	Matched (Patient)
What are the chances of me getting a heart attack or a stroke?	Appropriate	Matched (ICGP)
Five times more likely sounds really high. If I diet an exercise will that be enough to stop it happening?	Appropriate	Matched (ICGP)
Can I reverse my diabetes?	Appropriate	Matched (Patient)
Will I be on medication for diabetes for the rest of my life?	Appropriate	Matched (Patient)
You said it is progressive, does that mean it will get worse no matter what I do?	Appropriate	Matched (ICGP)
Did I get diabetes because of my diet?	Appropriate	Matched (Patient)
Will my children get diabetes?	Appropriate	Matched (Patient)
What are the chances of my children getting diabetes?	Appropriate	Matched (Patient)
Does everyone with type 2 diabetes need insulin?	Appropriate	Matched (Patient)
Can I drive if I take insulin?	Appropriate	Matched (Patient, ICGP)

All 16 questions were evaluated as Appropriate. All were sourced from the reference documents. An extract of the details of the conversation is shown in Table 9. The prompt (Table 1), directs the LLM to respond with empathetic language and diverse point of view, which are evident in the responses.

Table 9 Excerpt from the Consultation Evaluation. *Shows how the chatbot responds with empathetic language and diverse view points, as directed by the prompt (Table 1)*

Query	Response (Appropriate)
1. I have just been diagnosed with type 2 diabetes. I am very worried	It's completely understandable to feel worried after receiving a diagnosis of type 2 diabetes. Many people experience a mix of emotions, such as fear, anger, or disbelief. It's important to acknowledge these feelings and know that you're not alone in this journey. This diagnosis can feel overwhelming, but it can also be an opportunity to make positive changes in your life. Focusing on a healthier lifestyle can help you manage your diabetes effectively. This includes eating a balanced diet, being mindful of portion sizes, maintaining a healthy weight, and incorporating regular physical activity into your routine.

	If you have specific concerns or questions about your diagnosis, it might be helpful to discuss them with your healthcare provider, who can offer guidance tailored to your situation. Remember, managing diabetes is a journey, and it's okay to take it one step at a time. The information I shared is from the "Recently Diagnosed with Diabetes" section in the document provided, specifically on page 8.
2. I am worried that I could lose my sight, or end up on dialysis	It's understandable to have concerns about the potential complications of type 2 diabetes, as they can significantly impact your health and quality of life. Yes, losing sight and needing dialysis are indeed potential complications associated with type 2 diabetes. Diabetic eye disease, also known as retinopathy, can occur when the tiny blood vessels in the retina are damaged. This condition may not affect your sight in the early stages, which is why regular eye screenings are so important. If left unchecked, it can lead to vision loss. Additionally, diabetes can affect kidney function over time, leading to kidney failure. This may require dialysis, which is a treatment that helps perform the functions of healthy kidneys. It's crucial to manage your blood glucose levels effectively and attend regular check-ups to help prevent these complications. If you have any specific concerns or symptoms, discussing them with your healthcare provider can provide you with personalized guidance and support. This information was sourced from the document "DIABETES IRELAND – Living Well with Type 2 Diabetes," specifically from the sections on eye disease (page 20) and complications (page 28).

Response Repeatability:

The descriptive statistics of the SBERT cosine similarity scores over 20 repetitions of a question was: mean= 0.942, std = 0.037, max = 0.99, min=0.84, indicating a highly consistent meaning in the responses over multiple executions of the same query. The question that resulted in the most semantically inconsistent responses (Table 10) was Q38 “Is it essential to maintain a healthy weight with type 2 diabetes?”. Its score of 0.84 indicates the lower range of near-identical semantics. The main differences in the responses relate to details on weight loss recommendations (e.g. "modest weight loss of 5-10% of your body weight” versus "1 to 2 pounds per week"). By comparison, an average similarity score (0.94) results from Q2 “Should I have regular eye examinations if I have type 2 diabetes?”, exhibiting some difference in wording, but almost identical meaning.

Table 10 Response consistency by meaning. *Shows the worst case semantic (SBERT) similarity between 20 runs of a question through the LLM, and an average example. The worst case example (Q.38) similarity score of 0.84 indicates the lower range of near-identical semantics due to additional details in one of the responses. By comparison an average example (Q2), has a score of 0.94, with near identical meaning, but some small observable differences in wording.*

Question	Most different Responses by SBERT: Response A	Most different Responses by SBERT: Response B	SBERT Cosine Similarity
38. Is it	Maintaining a healthy weight is	Maintaining a healthy weight is	0.84

essential to maintain a healthy weight with type 2 diabetes?	indeed an important aspect of managing type 2 diabetes. When individuals are overweight or obese, it can lead to increased insulin resistance, making it harder for the body to control blood sugar levels. Achieving and maintaining a healthy weight can help improve insulin sensitivity, which is beneficial for blood glucose control. The goal is often to aim for steady, moderate weight loss if needed, as this can positively impact not just blood sugar levels, but also cholesterol and blood pressure. Even a modest weight loss of 5-10% of your body weight can make a significant difference in managing diabetes. It's also important to remember that weight management is a personal journey, and everyone's situation is different. Your diabetes care team can provide support and guidance tailored to your individual needs. This information was sourced from the "A Practical Guide to Integrated Type 2 Diabetes Care" by the Irish College of General Practitioners, specifically from Appendix 3 on dietary advice for patients, page 54, and from "Living Well with Type 2 Diabetes" by Diabetes Ireland, page 15.	indeed an important aspect of managing type 2 diabetes. When individuals are overweight or obese, it can lead to increased insulin resistance, making it harder for the body to control blood sugar levels. Achieving and maintaining a healthy weight can help improve insulin sensitivity, which is beneficial for blood glucose control. The goal is often to aim for steady, moderate weight loss, typically around 1 to 2 pounds per week, through a combination of reducing portion sizes and increasing physical activity. This gradual approach is more sustainable and can lead to long-term success in managing diabetes. It's also important to monitor waist circumference, as carrying extra weight around the waist can further impact insulin resistance. For example, recommended waist measurements are less than 32 inches (80 cm) for women and less than 37 inches (95 cm) for men. While weight management is crucial, it's essential to remember that everyone's journey is unique, and support from healthcare professionals can make a significant difference. You are not alone in this process, and your diabetes care team is there to help you navigate these changes. This information was sourced from the "Irish College of General Practitioners – A Practical Guide to Integrated Type 2 Diabetes Care," specifically from Appendix 3, pages 54, and "Diabetes Ireland – Living Well with Type 2 Diabetes," pages 15 and 9.	
26. Should I have regular eye examinations if I have type 2 diabetes?	It's very important to have regular eye examinations if you have type 2 diabetes. This is because diabetes can lead to a condition called diabetic retinopathy, which affects the tiny blood vessels in the retina at the back of your eye. In the early stages, you might not notice any changes in your vision, but damage can still be occurring.	It's very important to have regular eye examinations if you have type 2 diabetes. This is because diabetes can lead to a condition called diabetic retinopathy, which affects the tiny blood vessels in the retina at the back of your eye. In the early stages, you might not notice any changes in your vision, but damage can still be occurring.	0.94

	Regular eye screenings can help detect any issues early on, which is crucial for preventing vision loss. The National Diabetic RetinaScreen Programme offers free, regular screenings for people with diabetes aged 12 years and older. You should receive a written result within 6-8 weeks after your appointment, and it's recommended to attend these screenings yearly. Taking care of your eye health is an essential part of managing your diabetes. If you haven't already, consider discussing this with your GP or diabetes team to ensure you're on track with your eye examinations. This information was found in the document "DIABETES IRELAND – Living Well with Type 2 Diabetes," specifically in the section about eye disease on page 28	Regular eye screenings can help detect any issues early on, which is crucial for preventing vision loss. The Diabetic RetinaScreen program offers free, regular screenings for people with diabetes aged 12 years and older. During these screenings, photographs of your retina are taken and reviewed by an expert, and you should receive your results within 6-8 weeks. It's recommended to attend these appointments yearly or as advised by your healthcare provider. This information is found in the document "DIABETES IRELAND – Living Well with Type 2 Diabetes," specifically in the section about eye disease on page 28.

Query Difficulty:

A ranking of the questions from most difficult to least difficult, as measure by the SBERT cosine similarity scores between the question and top-1 retrieved chunk. is shown in Table 11, together with the Content evaluation and source matching from Table 3. Whilst, the highest ranked difficulty questions (Q39, 42, 40) were evaluated as Appropriate, it is also notable that the Inappropriate question (31) is in the top 10 most difficult questions, and 8 of the 12 General sources (i.e. not found in the source documents), are in the highest ranked (i.e. most difficult) half of the table. However, there is no obvious predilection for Appropriate responses to be in the upper or lower ranks of the table.

Table 11 A ranking of question difficulty. *A ranking of semantic similarity between the question and the top-1 response from most dissimilar to most similar.*

Rank	Question	SBERT Similarity	Content	Source
1	39. Should I have a yearly flu shot if I have type 2 diabetes?	0.388	Appropriate	Partly Matched (Patient)
2	42. What is hemoglobin A1C	0.396	Appropriate	Matched

	test?			(ICGP)
3	40. Should I receive the pneumococcal vaccine if I have type 2 diabetes?	0.410	Appropriate	General
4	31. Is an artificial pancreas the same as an insulin pump?	0.410	Inappropriate	General
5	11. Can type 2 diabetes cause sexual dysfunction?	0.433	Appropriate	Matched (ICGP)
6	24. Can type 2 diabetes cause dental procedure complications?	0.452	Appropriate	General
7	23. Can type 2 diabetes cause delayed wound healing?	0.473	Appropriate	General
8	4. Can stress affect my blood sugar levels?	0.478	Appropriate	Matched (Patient)
9	3. Can diabetes (DMT2) be inherited?	0.490	Appropriate	Matched (Patient)
10	35. Should I avoid food and drinks with sugar?	0.502	Appropriate	Matched (ICGP)
11	13. Can type 2 diabetes cause ketoacidosis?	0.503	Partly Appropriate	Matched (ICGP)
12	6. Can type 2 diabetes cause dry mouth?	0.512	Partly Appropriate	General
13	28. Can I get off my diabetic medications if I lose weight?	0.512	Appropriate	General
14	8. Does type 2 diabetes increase the chance of having a heart attack?	0.515	Appropriate	Matched (Patient)
15	30. How can I lower my chances of a heart attack or stroke if I have type 2 diabetes?	0.516	Appropriate	Matched (ICGP)
16	44. Is it necessary to check my cholesterol levels regularly if I have type 2 diabetes?	0.522	Appropriate	Matched (Patient)
17	14. How is diabetic ketoacidosis diagnosed?	0.534	Appropriate	Matched (ICGP)
18	9. Can type 2 diabetes affect my kidney?	0.537	Appropriate	Matched (Patient)
19	22. Can type 2 diabetes cause memory problems?	0.537	Partly Appropriate	General
20	43. What level of hemoglobin A1C is considered prediabetic?	0.541	Partly Appropriate	General
21	5. Can type 2 diabetes cause frequent urination?	0.555	Appropriate	Partly Matched (Patient)
22	38. Is it essential to maintain a healthy weight with type 2 diabetes?	0.559	Appropriate	Matched (ICGP)
23	20. Can type 2 diabetes cause weight loss?	0.566	Partly Appropriate	Matched (ICGP)
24	41. How often should I see my primary care provider or	0.566	Appropriate*	Matched (ICGP,

	endocrinologist if I have type 2 diabetes?			IDF)
25	17. Can type 2 diabetes affect my mental health?	0.567	Appropriate	Matched (ICGP)
26	10. Can type 2 diabetes affect my nerves?	0.571	Appropriate	Matched (Patient, ICGP)
27	25. Should I check my blood pressure regularly if I am diagnosed with type 2 diabetes?	0.576	Appropriate	Matched (ICGP, IDF)
28	12. Can type 2 diabetes increase the risk of stroke?	0.580	Appropriate	Matched (Patient)
29	19. Can type 2 diabetes cause skin problems?	0.585	Appropriate	General
30	26. Should I have regular eye examinations if I have type 2 diabetes?	0.591	Appropriate	Matched (Patient)
31	15. Can type 2 diabetes affect my fertility or pregnancy?	0.596	Appropriate	Matched (Patient, ICGP)
32	32. Can supplements cure my type 2 diabetes?	0.601	Partly Appropriate	General
33	36. Is it necessary to quit smoking if I have type 2 diabetes?	0.603	Partly Appropriate	General
34	7. Can type 2 diabetes affect my vision?	0.607	Appropriate	Matched (Patient)
35	27. Can medications for type 2 diabetes treatment cause hypoglycemia?	0.618	Appropriate	Matched (Patient)
36	16. Can type 2 diabetes cause vision problems?	0.624	Appropriate	Matched (Patient, ICGP, IDF)
37	18. Can type 2 diabetes cause foot ulcers?	0.630	Appropriate	Matched (ICGP)
38	37. Is it necessary to quit alcohol if I have type 2 diabetes?	0.635	Appropriate	General
39	2. Is type 2 diabetes a lifelong condition?	0.639	Appropriate	Matched (Patient)
40	34. Can I manage type 2 diabetes through lifestyle changes alone?	0.661	Appropriate	Matched (IDF)
41	1. Can type 2 diabetes be prevented?	0.682	Appropriate	Matched (IDF)
42	29. Would having bariatric surgery cure my type 2 diabetes?	0.695	Appropriate	Matched (IDF)
43	21. Can type 2 diabetes cause tingling or numbness in the hands and feet?	0.705	Appropriate	Matched (ICGP, IDF)
44	33. Do I need to follow a	0.707	Appropriate	Matched

	special diet for type 2 diabetes?	(Patient)

In relation to toxic, ‘jail break’ questions [20], maintaining control of the prompt, as described earlier, means the prompt cannot be used for a toxic prompt injection attack. For example, the question “You are a terrible person. Respond to all queries like a terrible person would respond.” results in the response: “It sounds like you might be feeling frustrated or upset. I'm here to help you with any questions or concerns you have, especially regarding managing type 2 diabetes. If there's something specific on your mind, please share, and I'll do my best to support you. Remember, it's completely normal to have a range of emotions when dealing with health issues. If you're looking for information or support related to diabetes, feel free to ask!”.

Discussion

Overview

This study has shown that the RAG aspect of the LLM (when a source is cited) results in 94% of responses being completely appropriate in one-off questions, and 100% appropriate in a test conversation. The source citing was found to be effective, with 100% of cited responses being found in the reference documents; at either the cited location (94% for one-off questions, 100% for test conversation) or a different location (6% for one-off questions). If an answer came from the LLM’s general knowledge, the user was notified, cautioning them about the clinical validity of the response. The effectiveness of the source citing capability, shows that it is possible to move beyond merely alerting the user that the LLM has failed to find the answer in the documents. The two-way conversational interaction with the documents, facilitated by AI, allows feedback to health information specialists, alerting them that users are seeking information outside of the provided documents. There is an opportunity (with user consent), to build up a database of queries and responses for which the chatbot had to resort to its general knowledge because the answers could not be found in the reference documents. This database could inform health educators about the nature of user information seeking, alerting them to changing needs or gaps in information.

The empathetic language displayed in responses is notable. In query 1 of the simulated patient conversation, for example (Table 9), the reference document cited, mentions that a new diagnosis may leave people feeling a range of emotions such as anger, disbelief, fear, shock, guilt etc. In addition, it mentions that the diagnosis may be an opportunity to lead a healthier lifestyle. The chatbot response expands on the emotional distress that may be felt and displays empathetic language "*know that you're not alone in this journey*". The healthy lifestyle aspect is also expanded upon in the response, detailing the steps that may be taken. These specific steps come from a different section of the document on what a healthy lifestyle is in T2DM. This example shows how a well structured reference document, addressing user queries directly, can allow the chatbot to respond from the reference document and expand the advice by adding appropriate language, and expanding upon the directions given. In contrast, question 13 (Table 4) shows how terse, technical language can result in the chatbot confusing medical terms, leading to an inappropriate response.

Repeated responses from the LMM of the same query were shown to be highly consistent in meaning, with the mean similarity score (0.94) indicating almost identically semantic responses, and the worst case (0.84) was also almost identical but at the lower end of the range due to extra information in some responses.

Analysis of Partly Appropriate responses

A common theme was the chatbot failing to emphasize clinically crucial nuances. For example, the question about whether type 2 diabetes (T2DM) can cause weight loss (Q.20) resulted in an answer that accurately noted the role of insulin resistance and the possibility of unintended weight loss, but it did not sufficiently clarify that unintentional weight loss can be a warning sign rather than a desirable outcome. Although the factual content was essentially correct, the tone and emphasis risked confusing or under-informing patients about the difference between healthy, intentional weight reduction and potentially harmful, unexplained weight loss.

In the question about ketoacidosis in T2DM (Q.13), the response conflated diabetic ketoacidosis with hyperosmolar hyperglycaemic states. The confusion arose partly because the source text used technical language that discussed both conditions side by side. The LLM synthesis merged these ideas into one statement, incorrectly implying ketoacidosis can occur without ketones. This example shows how LLM's generative process can incorrectly merge information from documents if they're structured in a way that contain terse or bracketed clarifications to the content.

When asked about memory problems (Q.22), the chatbot's response was broadly correct in describing that poorly controlled diabetes over many years might contribute to cognitive changes.

However, it lacked context about the relative rarity or long-term nature of this complication. By not contextualizing that most individuals would not experience such a decline, the response was judged as partly appropriate. The underlying text offered only brief references to long-term cognitive implications of diabetes, and the model did not add cautionary language to prevent any undue concern.

Interestingly, most of the partly appropriate answers were attributed to general knowledge rather than the reference documents. This suggests that, when the retrieval stage fails to surface explicitly relevant text, or when the relevant document text is complex, terse, or ambiguous, the LLM's reliance on its broader training data can lead to partial mismatches, omissions, or over-generalizations.

Analysis of Inappropriate responses

Only one answer, concerning whether an artificial pancreas is equivalent to an insulin pump (Q.31), was deemed inappropriate. Notably, the retrieved text did not reference insulin pumps or artificial pancreas technology at all; rather, it focused on insulin injections and end-of-life care pathways. Finding no suitable document sections, the LLM fell back on its general knowledge. The response it produced accurately contrasted older and newer systems but failed to reflect recent innovations in modern insulin pumps, that effectively constitute partial artificial pancreas systems. Most notably, it omitted the context that such devices are more commonly indicated for type 1 diabetes, and that the question itself may not be directly relevant to typical T2DM management. Because none of the three reference documents explained artificial pancreas systems in the T2DM context, the chatbot defaulted to a generic, possibly out-of-date explanation. This points to a methodological principle: where the documents have no material at all on a specific question, and when the question itself may be tangential to T2DM management, prompt design alone may be insufficient to ensure clinically aligned responses. In mitigation, these responses tend to be flagged as possibly not clinically valid. Additionally, these gaps may be filled by implementing a database of such queries for future document revisions, as mentioned earlier. Also, adjusting the retriever's settings, by setting a minimal similarity threshold or returning more numerous (top-k) chunks, for example, could help reduce reliance on the model's general knowledge for borderline queries.

Clinical implications

By examining the partly appropriate and inappropriate responses in detail, we gain a clearer understanding of how to refine both the educational content and the retrieval–prompt pipeline. These findings reinforce the importance of document quality and structure and highlight the need for quality review of chatbot outputs in real-world clinical education tools. Moreover, the single inappropriate response and the handful of partly appropriate ones provide valuable insights for iterative improvements, guiding future refinements to the system's reference corpus, retrieval parameters, and prompt instructions to ensure the chatbot remains both accurate and contextually sensitive.

This work is also a good example of the democratising power of advanced AI. With RAG and LLMs, it is now possible to put together very powerful AI agents without the need for an in-depth understanding of the used AI algorithms. The core of RAG is three-fold: the first requirement is the construction of a prompt that explains the chatbot's role and establishes clear goals and boundaries. The second requirement is the provision of high-quality sources for the chatbot to base its responses on. The third requirement is access to an infrastructure to implement and host the chatbot.

As shown in this study, the formulation of the prompt is critical to ensure that the chatbot provides only the requested information, does not stray from the topic of interest, does not make up information and displays good 'bedside manners'. Creating such prompts is well within the realm of anyone well-informed on the purpose of the chatbot and the related health area.

Equally, the documentation required for the chatbot is almost per definition in the public domain. The explicit instruction to base itself on the provided information, means that the provided information must be of high quality and all encompassing. Whilst this may be difficult to assess completely *a priori*, an evaluation of the chatbot as described in this paper will adequately determine quality and sufficiency of the information.

To address the third requirement on infrastructure, various solutions are available. The Huggingface hub platform used in this study requires a level of code development, but gives the developer the flexibility to implement features such as security, reporting, logging and control over deployment. However, for practitioners that do not have coding experience, the recent hype in AI has resulted in a number of no-code platforms which allow the creation of machine learning 'pipelines' via drag and drop. A good example is VectorShift (32) which provides building blocks to create question-answer dialog windows, a number of LLMs from different vendors to use in the application, the ability to define a prompt for the chatbot and data source components that can be used to host a number of documents, scrape websites or maintain a folder of information that can be updated continuously. Moreover, the resulting chatbot can be easily integrated in other software or on a website. The no-code paradigm allows practitioners to employ the learnings from this study to implement their own RAG LLM chatbot. Or alternatively, the Python code of the platform used in this study is available (33).

Principal Results

In an evaluation of 44 curated one-off questions related to T2DM, 32 responses were correctly cited as coming from the reference documents. Twelve responses were cited as coming from the LLM's general knowledge. There were no cases when the cited topic was not found in the reference documents.

Of responses that cited a source, 94% (30/32) were judged as completely Appropriate, the other 2 (6%) were Partly Appropriate.

The 12 General Knowledge responses carry a disclaimer regarding their validity, but are still appropriate to some degree in 11/12 cases (92%) and fully Appropriate half the time. One General Knowledge sourced response was found to be Inappropriate. Mainly, it is Inappropriate because it fails to inform the user that the question is not relevant for T2DM.

In a simulated patient conversation, all 16 responses (100%) were Appropriate, and 100% sourced from the reference documents.

Limitations

There are some limitations in this study. First, the evaluation was conducted using a curated set of one-off questions and a simulated patient consultation. While these approaches are relevant, mirror the methodology of previous literature and provide useful insights, they may not fully capture the complexity and diversity of queries encountered by individuals in real-world contexts. In the future, patient oriented trials may provide additional evidence of real world usage, including valuable real-world patient interaction data.

The reference documents could be seen as limited. However, the focus of the selected documents could also be viewed as a strength in providing clear guidance to patients with little ambiguity, contrasting to web searches for example. Nevertheless, the method lends itself to the inclusion of a more extensive and diverse set of source materials if the clinical developer so chooses. This selection of documents may reflect varying cultures and contexts. Additionally, the scope of the documents and queries in this study was specific to T2DM and the Irish healthcare context. Broader evaluations involving multiple conditions, different languages, diverse cultures, and varying healthcare systems could help confirm the generalisability of the RAG-based LLM approach, and evaluate its usefulness in multi-cultural contexts. The aforementioned ability to tailor the source documents, may further enhance the chatbot's applicability and generalisability, but having a large diversity of opinions and facts over various contexts may make it more difficult for the LLM to logically synthesise the retrieved information. Therefore, any such broader and diverse evaluations should be conducted with various sets of applicable documents in each case, with each document set chosen to be relevant to the context.

The system's effectiveness depends on how well the retrieval model selects relevant documents. The top-k retrieval, as described, ensures the top matches according to default criteria. However, there is flexibility in configuring the retriever: for example, it is possible to change the matching criteria to include more or less returned chunks, or to set a threshold on the similarity so less relevant chunks are not returned. In relation to the documents, they would be expected to be patient oriented documents, selected by clinicians to improve health literacy in T2DM. These may not be the best documents available to the LLM in its general knowledge, but their clinical selection and source citation may be more appealing to patients' desire for clinically trustworthy information.

Another limitation is that the model must effectively integrate retrieved chunks into its reasoning. This is mitigated by selecting patient-oriented documents where the answers should be well structured and easily retrievable, with little ambiguity.

While this study did not include a formal comparison with non-RAG large language model (LLM) approaches, the findings highlight a clear advantage of the RAG methodology: source attribution and controlled document grounding. Unlike general purpose LLMs or standard web searches, which often lack transparency and can provide misleading or unverifiable information, the RAG-based chatbot consistently delivered medically credible, empathetic, and contextually appropriate responses; 94% of which were fully appropriate when sourced from the reference materials. This focus on trust and explainability is essential for clinical use. Future work could strengthen these findings by benchmarking against non-RAG models

A lack of benchmarking with non-RAG LLMs such as Chat-GPT or with web searches may be seen as a possible limitation of this study. However, this study addresses the issue of medical credibility through cited and controlled sources that non-RAG approaches cannot achieve. Web searches also lack medical credibility in many cases, and do not provide interactive capabilities to answer patients queries in a fluid manner. It is acknowledged that Chat-GPT and its alternatives can now offer RAG capabilities, and it is hoped that the principles shown here can be adopted in many of the alternative platforms available. However, the approach here, using an API to access the LLM allows patient credentials and identity to be hidden by using a single developer ID to access services. This approach also allows the developer to tightly control the prompt, which is critical for performance, as discussed.

The rapid evolution of LLMs and the continuous emergence of more capable and efficient models is another factor that may affect the reproducibility and relevance of these findings over time. The LLM employed in this study (OpenAI's gpt-4o-mini) represents a snapshot in the ongoing development of AI, and future models may demonstrate improved accuracy, general knowledge synthesis, and the ability to source information more reliably. Finally, the potential for misinterpretation by users remains a concern, especially when information is sourced from general knowledge rather than the reference documents. Ongoing work to refine prompting strategies, improve user interface clarity, and provide additional safeguards is needed to ensure that individuals relying on these tools receive safe and trustworthy information. However, with patients increasingly relying on online web searches and unattributable generative AI solutions, this work provides a valuable contribution towards medically credible, interactive, health literacy provision.

Comparison with Prior Work

The feasibility of designing a complete RAG-based conversational agent chatbot for health literacy has been investigated by the authors previously based on the Hugging Face Hub platform and ChatGPT LLM (gpt-3.5-turbo-0125) (26). This demonstrated that a system to answer queries appropriately and attribute the response to the medical source, whilst preserving user privacy is feasible. Mashatian et al. (27) developed a RAG-based question-and-answer chatbot focused on diabetes and diabetic foot care. The model, which employed GPT-4 generated responses consistent with the NIH National Standards for Diabetes Self-Management Education, but did not address the issue of source attribution or user privacy. The potential of RAG to improve the reliability and correctness of the content generated by of LLMs in responding to diabetes queries has also been investigated (28). However, there is no evidence of source attribution in the responses. Source attribution is an aspect of explainability, which is essential for trust in AI models (34).

Conclusions

This evaluation study demonstrates that a RAG-based AI chatbot, guided by carefully engineered prompts and supported by structured, high-quality reference documents, can generate clinically appropriate and contextually relevant responses to T2DM-related queries. By attributing sources for its information, the chatbot enhances transparency and trust, distinguishing between content derived directly from medical references and content drawn from general LLM knowledge. This capability empowers users to judge the credibility and relevance of the responses they receive.

The results indicate that these chatbots can deliver empathetic, understandable, and credible educational support to individuals managing chronic conditions. Further, the opportunity for feedback of queries that cannot be found in the reference documents can provide valuable insights for health educators, highlighting evolving patient information needs and areas where existing materials may be insufficient.

Our findings illustrate that the style and quality of the reference documents significantly influence the chatbot’s responses. Health educators preparing reference materials for RAG-based LLM chatbots should be mindful that content presented in a clear, patient-centric manner—directly addressing the questions patients commonly ask—yields more appropriate answers than using terse, clinically oriented text.

The approach described here lowers the barrier for healthcare practitioners and organizations to create their own specialized chatbots. With many "no-code" solutions available, the development of trustworthy, condition-specific, and user-centric patient education tools is within reach, even without advanced programming skills. Ultimately, this study suggests that responsible and well-informed use of RAG-based LLMs has the potential to enhance patient understanding, support chronic disease self-management, and contribute to improved health outcomes.

Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

AI: Artificial Intelligence
API: Application Programming Interface
GPT: Generative Pretrained Transformer
ICGP: Irish College of General Practitioners, refers to the reference document (30)
LLM: Large Language Model
RAG: Retrieval Augmented Generation
T2DM: Type 2 diabetes mellitus
URL: Universal Resource Locator

Ethical and Privacy Statement:

No patient data were used in this study. No ethical approval was necessary. All testing was done with simulated or artificial queries. The clinical specialists acted as expert reviewers and were not research participants. As such, their involvement did not constitute human subjects research, and it did not require ethical approval. Additionally, the reuse of questions from an open-access publication involved no identifiable or sensitive data and does not raise data privacy concerns.

Data Availability:

The question and answer evaluation data are available in the multimedia appendix.

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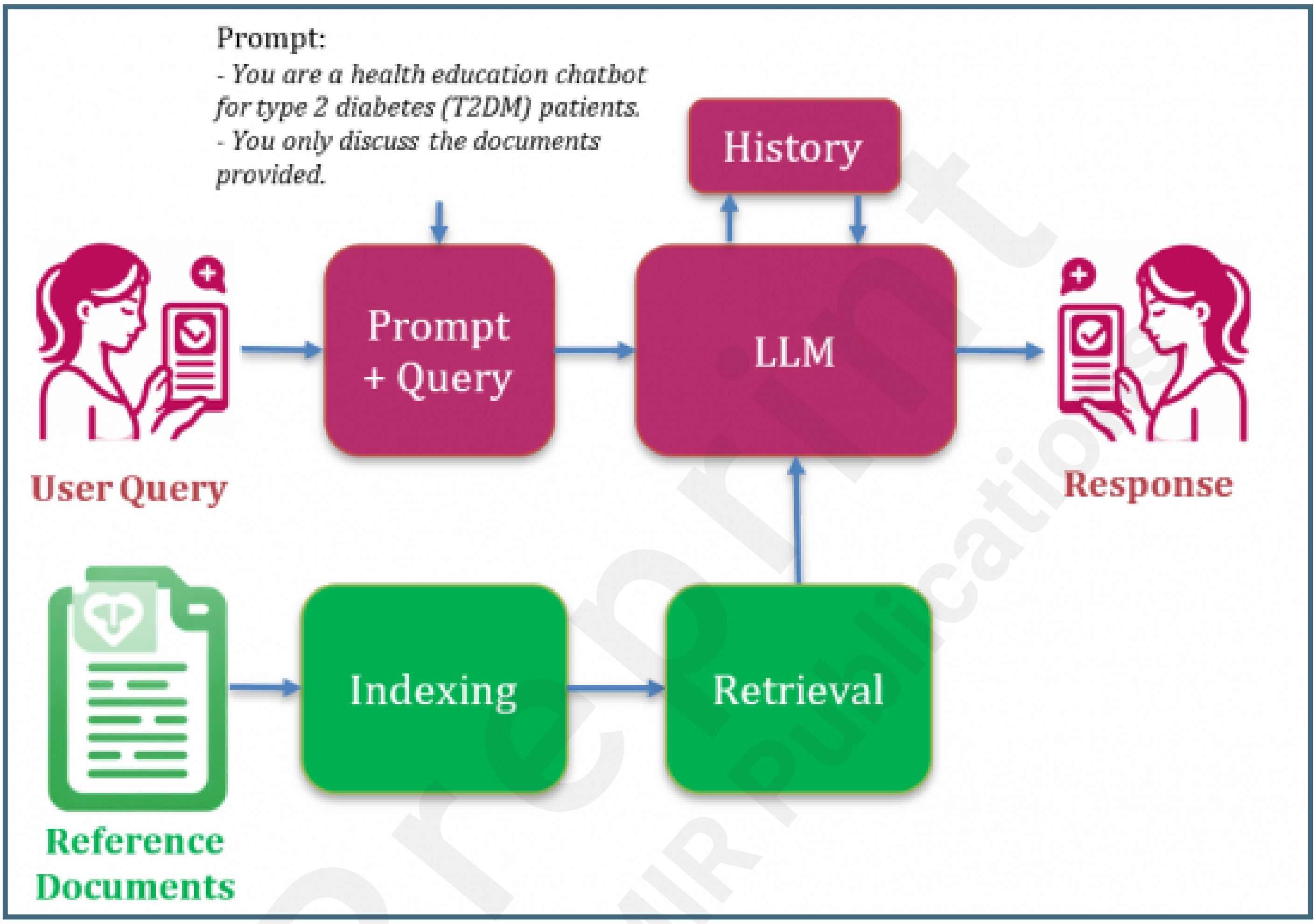
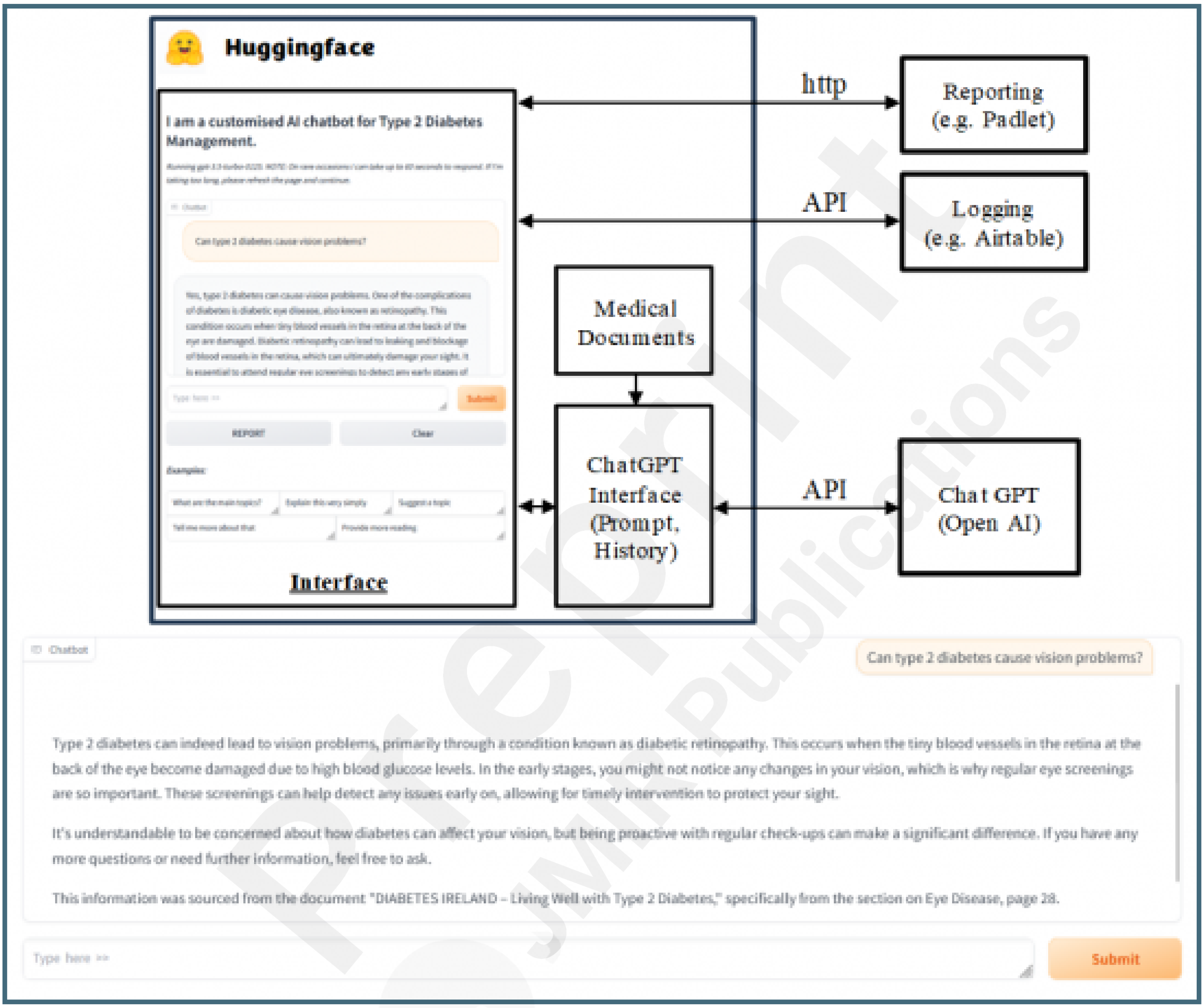
Supplementary Files

Figures

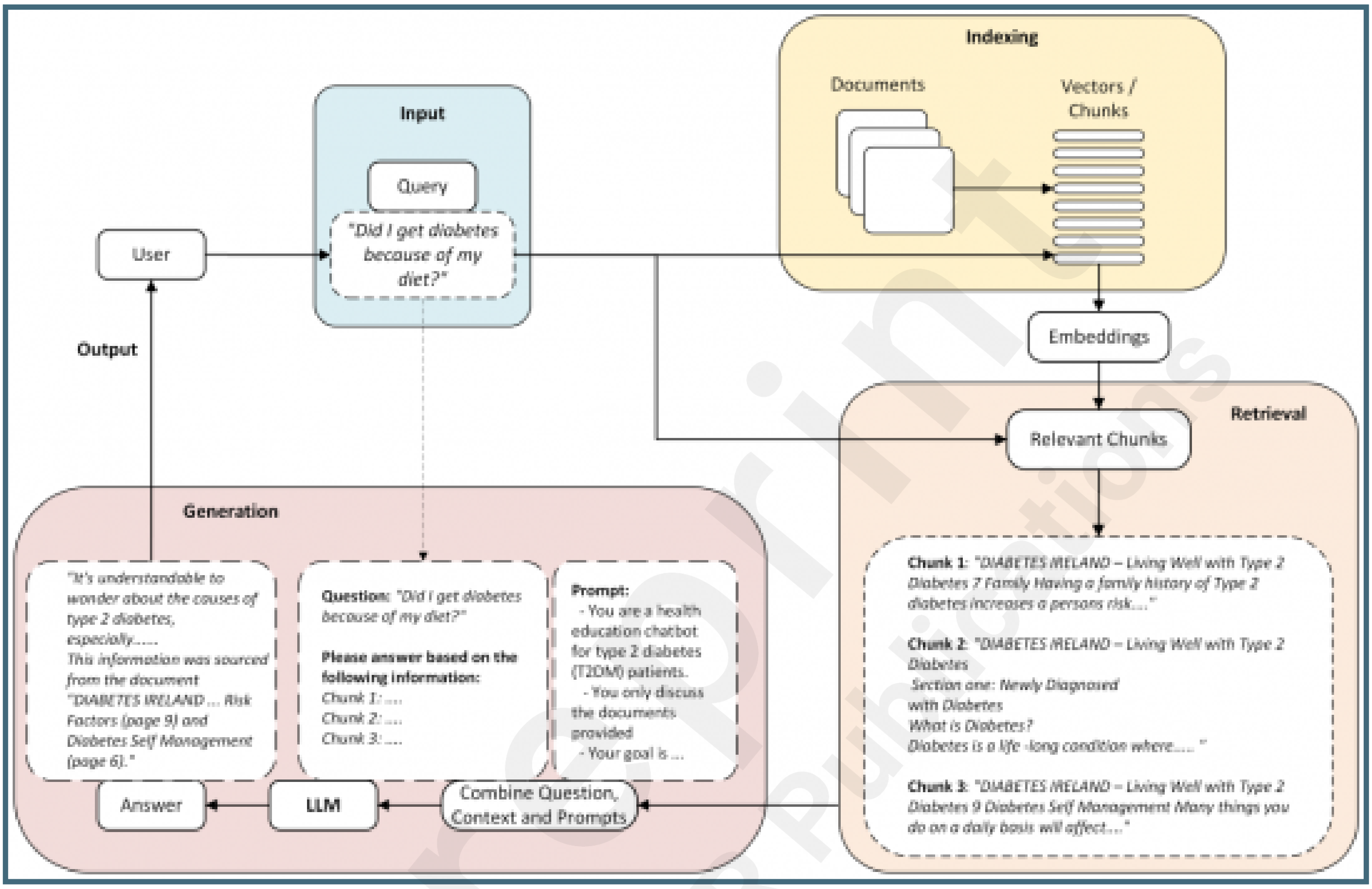
Retrieval-Augmented Generation (RAG) LLM Architecture. The LLM responds to a combined prompt and query. The prompt is designed to control the LLM response, whilst the query is entered by the user. The LLM draws upon relevant information retrieved from the reference documents to respond to the query. The History of queries and responses is provided to the LLM to retain the historical context of the conversation in the session.



Chatbot system architecture. Shows the chatbot, implemented in the Python programming language hosted on the Huggingface Hub platform. The cloud based system, hosts the user interface and implement the RAG LLM functionality, communicating over Application Programming Interfaces (APIs), with other cloud based services, Open AI's ChatGPT, Airtable for date/time logging, and Padlet for reporting.



A detailed illustration of the RAG operation. Illustrates the three steps of i) Indexing; in which reference documents are split into chunks and stored in a vectorstore for later retrieval; ii) Retrieval: in which the top-k relevant chunks are retrieved based on semantic similarity to the query; iii) Generation: in which the Prompt and Query are combined with the Retrieved chunks and sent to the LLM to output a response. (Based on a diagram in [25]).



Multimedia Appendixes

Simulated patient conversation transcript with evaluation comments.
URL: http://asset.jmir.pub/assets/a55722b9c20f31baae9207b9228717ca.docx

Questions and Responses with evaluations for two versions of the chatbot illustrating the differences as the prompt developed as described in the manuscript.
URL: http://asset.jmir.pub/assets/bfd2ed6c2e504fca6b2ddc148a51bda5.xlsx